

Integrable Systems — Exercise Sheet 11

The Lieb–Liniger gas

Variant: Questions only

Conventions, as in Lecture 11: units $\hbar = 2m = 1$; $H = -\sum_j \partial_j^2 + 2c \sum_{i < j} \delta(x_i - x_j)$ for N bosons on a ring of length L , $c > 0$ unless stated; $S(k) = (k - ic)/(k + ic) = -e^{i\theta(k)}$ with $\theta(k) = 2 \arctan(k/c)$; the Bethe equations are $e^{ik_j L} = \prod_{l \neq j} \frac{k_j - k_l + ic}{k_j - k_l - ic}$, with $E = \sum_j k_j^2$. “Free fermions”, as in Lecture 11, are N free fermions on the ring with $\psi(\dots, x_j + L, \dots) = (-1)^{N-1} \psi(\dots, x_j, \dots)$: periodic for odd N , antiperiodic for even N .

Exercise 1 (the contact condition). Two bosons, with $X = \frac{1}{2}(x_1 + x_2)$ and $r = x_2 - x_1$. (i) Show that $\partial_1^2 + \partial_2^2 = \frac{1}{2}\partial_X^2 + 2\partial_r^2$ and $\partial_r = \frac{1}{2}(\partial_2 - \partial_1)$. (ii) Integrate $H\psi = E\psi$ over $-\epsilon < r < \epsilon$ and let $\epsilon \rightarrow 0$, to show that $\partial_r \psi$ jumps by $c\psi$ at $r = 0$. Why must ψ itself be continuous? (iii) For bosons ψ is even in r . Deduce the contact condition $(\partial_2 - \partial_1 - c)\psi = 0$ at $x_2 = x_1$, approached from $x_1 < x_2$.

Exercise 2 (the two-body S-matrix). (i) On $x_1 < x_2$ take $\psi = A_{12}e^{i(k_1 x_1 + k_2 x_2)} + A_{21}e^{i(k_2 x_1 + k_1 x_2)}$. Show that the contact condition holds exactly when $A_{21}/A_{12} = S(k_1 - k_2)$. (ii) Show that $S = -e^{i\theta}$ with $\theta'(k) = 2c/(k^2 + c^2) > 0$, that $S(0) = -1$, and find the limits of S as $c \rightarrow \infty$ and as $c \rightarrow 0$, at fixed $k \neq 0$. What do they describe? (iii) For two *distinguishable* particles with the same interaction, let the relative wave $e^{-ikr/2}$, $k = k_1 - k_2 > 0$, come in from $r > 0$: take $\psi = e^{-ikr/2} + \rho e^{ikr/2}$ for $r > 0$ and $\psi = \tau e^{-ikr/2}$ for $r < 0$. Find the transmitted and reflected amplitudes τ and ρ , and show that $|\tau|^2 + |\rho|^2 = 1$ and $\tau + \rho = S(k)$.

Exercise 3 (two bosons on a ring). (i) Put $k_{1,2} = \frac{K}{2} \pm q$ with $K = 2\pi n/L$. Show that the Bethe equations reduce to $qL + \theta(2q) = \pi(n+1) \pmod{2\pi}$, and that $E = \frac{1}{2}K^2 + 2q^2$. (ii) For $K = 0$ the ground state has $qL + 2 \arctan(2q/c) = \pi$ with $q > 0$. Solve it for $L = c = 1$, and find its limits as $c \rightarrow \infty$ and as $c \rightarrow 0$ at fixed L .

Exercise 4 (real roots, and the attractive gas). (i) For $N = 2$ and $K = 0$, look for imaginary roots $k_1 = i\kappa$, $k_2 = -i\kappa$ with $\kappa > 0$. Show that the Bethe equations become $e^{-\kappa L} = \frac{2\kappa+c}{2\kappa-c}$. (ii) Show that for $c > 0$ it has no solution, as Lecture 11’s proposition on real roots says. (iii) For $c < 0$ show that it has a solution with $\kappa > |c|/2$, and that $\kappa \rightarrow |c|/2$ as $L \rightarrow \infty$: a bound pair, the case the lecture excludes. Solve it for $c = -1$, $L = 10$.

Exercise 5 (the logarithmic form, and the Yang–Yang function). (i) For real roots, take logarithms of the Bethe equations to get $k_j L + \sum_{l=1}^N \theta(k_j - k_l) = 2\pi I_j$ with $I_j \in \mathbb{Z} + \frac{N-1}{2}$, and show $K = \frac{2\pi}{L} \sum_j I_j$. (ii) With $\Theta(x) = \int_0^x \theta$, let $B(k) = \frac{L}{2} \sum_j k_j^2 - 2\pi \sum_j I_j k_j + \frac{1}{2} \sum_{j,l} \Theta(k_j - k_l)$. Show that the equations of (i) are $\partial B / \partial k_j = 0$, and that the Hessian of B is $L \mathbb{1} + \sum_{j < l} \theta'(k_j - k_l)(e_j - e_l)(e_j - e_l)^T$. Deduce that the solution is unique. (iii) Show that $I_j > I_l$ implies $k_j > k_l$.

Exercise 6 (strong and weak coupling, and five bosons). Take the ground state, $I_j = -\frac{N-1}{2}, \dots, \frac{N-1}{2}$, whose roots are symmetric, $\sum_j k_j = 0$. (i) *Strong coupling.* For $|k| \ll c$ use $\theta(k) \approx 2k/c$ to show $k_j \approx 2\pi I_j / (L + 2N/c)$: free fermions on a ring lengthened by $2N/c$. (ii) *Weak coupling.* If the roots’ differences are large compared with c , use $\theta(x) \approx \pi \operatorname{sgn} x - 2c/x$ to show that $k_j = \sqrt{2c/L} h_j$ with $h_j = \sum_{l \neq j} 1/(h_j - h_l)$. Check that the zeros of the Hermite polynomial H_3 , 0 and $\pm\sqrt{3/2}$, solve this for $N = 3$. (iii) (*Computational.*) For $N = 5$, $L = 5$,

solve the equations of Exercise 5 at $c = 0.1, 1$ and 10 , and compare with (i) at $c = 10$ and with (ii) at $c = 0.1$, using the zeros of H_5 .

Exercise 7 ★. *The displacement.* In this exercise the two bosons move on the infinite line, not on a ring, so that they meet once. Superpose two-boson solutions of Exercise 2 into packets: $\psi = \iint f_1(k_1)f_2(k_2) \left[e^{i(k_1x_1+k_2x_2)} + S(k_1-k_2)e^{i(k_2x_1+k_1x_2)} \right] e^{-i(k_1^2+k_2^2)t} dk_1dk_2$ on $x_1 < x_2$, with f_1, f_2 real, positive and narrow around $\bar{k}_1 > \bar{k}_2$ (a real envelope puts the packet's centre at the origin at $t = 0$; a phase e^{-ika} would shift it by a). Work to leading order in the width, following the central rays. (i) A packet sits where its phase is stationary in the momentum it carries. Show that before the collision ($t \rightarrow -\infty$) the particles are at $x_1 = 2\bar{k}_1t, x_2 = 2\bar{k}_2t$, and after it ($t \rightarrow +\infty$) at $x_2 = 2\bar{k}_1t - \theta'(\bar{k}_1 - \bar{k}_2)$ and $x_1 = 2\bar{k}_2t + \theta'(\bar{k}_1 - \bar{k}_2)$. (ii) In the centre-of-mass frame, $\bar{k}_1 = -\bar{k}_2 = \bar{k} > 0$, show that both particles come out delayed, and find the delay. What happens to it as $c \rightarrow \infty$ and as $c \rightarrow 0$?